ADITYA KUMAR

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LinkedIn | GitHub | Portfolio

TECHNICAL SKILLS

Programming Languages: C++, Java, JavaScript, TypeScript

Core Computer Science: Data Structures and Algorithms (DSA), Object-Oriented Programming (OOP), Database Management Systems (DBMS), Operating Systems, Computer Networks, System Design

Backend Development: Node.js, Express.js, REST APIs, WebSockets, Authentication, Authorization, Asynchronous Processing, Redis, BullMQ

Frontend Development: Next.js, React.js, HTML5, CSS3, Tailwind CSS, GSAP

Databases: SQL, PostgreSQL, PostGIS, MongoDB, MySQL, Prisma

Developer Tools: Git, GitHub, Docker, Postman, Vercel, VS Code

PROJECTS

Campunex — Campus Ride-Matching Platform

Live | GitHub | Aug 2026

Technologies: Next.js, Node.js, PostgreSQL, PostGIS, Redis, WebSockets

- Observed high campus traffic where students frequently commuted in separate vehicles despite traveling along similar routes, which motivated an end-to-end ride-matching platform to improve vehicle utilization.
- Designed the backend architecture around PostgreSQL/PostGIS for 500-meter geospatial matching and WebSockets for real-time GPS tracking and trip-state synchronization.
- Evaluated performance and reliability requirements to implement Redis-based geospatial caching and OTP authentication, enabling efficient matching and secure ride initiation and completion.

Devnoesis — CS Learning & Developer Intelligence Platform

Live | GitHub | Aug 2026

Technologies: Next.js, React.js, Node.js, Express.js, TypeScript, PostgreSQL, Prisma, Redis, BullMQ

- Identified the need for a unified view of developer progress and built a platform that aggregates coding activity, CS knowledge, and preparation metrics into one system.
- Architected an asynchronous data pipeline using Redis and BullMQ workers to synchronize multiple external APIs with rate limiting, retry handling, and background job processing.
- Modeled a CS knowledge graph using DAG-based prerequisite relationships and applied the SM-2 spaced-repetition algorithm to generate personalized revision schedules.
- Translated coding activity, CS coverage, repository consistency, and preparation metrics into a Placement Readiness Score (0–100) through a unified Next.js dashboard.

AI FanFic — AI-Powered Story Generation Platform

Live | GitHub | Mar 2026

Technologies: React.js, JavaScript, Tailwind CSS, Gemini API, Web Speech API, Framer Motion

- Explored generative AI for interactive storytelling and turned the concept into a 3 mode platform supporting text chat, voice-based storytelling, and image generation using the Google Gemini API.
- Discovered the application around streaming AI responses and Web Speech API voice input to support interactive, low-friction user interactions during story generation.
- Shipped a responsive React.js product on Vercel used by 5 test users, with configurable story generation, theme switching, and Framer Motion interactions across desktop and mobile experiences.

TRAINING

Centre for Professional Enhancement, Lovely Professional University

Jun 2026 – Jul 2026

Advanced Backend Development with Node.js, Express and MongoDB | Certificate

- Completed structured training in Node.js, Express.js, MongoDB, server-side programming, backend application development, and REST API development.
- Gained practical experience with Express.js application architecture and MongoDB database integration.

CERTIFICATIONS

Data Structures and Algorithms in Java — iamneo, an NIIT Venture | Certificate — Jun 2026

Database Management System Part 1 — Infosys | Certificate — Aug 2026

Generative AI — NASSCOM Skill Development Program | Certificate — Mar 2026

EDUCATION

Lovely Professional University, Phagwara, Punjab

Aug 2025 – Present

B.Tech — Computer Science and Engineering

Expected Graduation: 2028 | CGPA: 8.07/10

Vidya Memorial Institute of Technology, Ranchi, Jharkhand

Jul 2023 – Jul 2025

Diploma — Mechanical Engineering | Percentage: 83%